

Assignment 1: Data Validation

Aim: To demonstrate data validation by building a login and sign-up form that checks every user input and shows clear error messages before the data is accepted.

Theory

Data validation is the process of checking that input is complete, well-formed and within expected limits before the application uses it. Unvalidated input is the root of many attacks, including SQL injection, cross-site scripting and broken business logic, so it should be enforced in the browser for usability and again on the server for security.

- **Presence check:** required fields are not left empty.
- **Format check:** the value matches a pattern, such as an email address or phone number.
- **Range / length check:** usernames and passwords respect minimum and maximum lengths; age is within an allowed range.
- **Consistency check:** related fields agree, such as password and confirm password.
- **Consent check:** a mandatory checkbox, such as accepting the terms, is ticked.

Validation rules implemented

Field	Rule	Error shown on failure
Username	3-16 characters: letters, digits, underscore	Username must be 3-16 letters, digits or underscores
Email	Regular expression for name@domain.tld	Enter a valid email address
Phone	10 digits, first digit 6-9	Phone must be 10 digits and start with 6-9
Date of birth	Required, in the past, age at least 13	You must be at least 13 years old
Password	8+ characters with upper, lower, digit and symbol	Password must be at least 8 characters, and so on
Confirm password	Identical to password	Passwords do not match
Terms	Checkbox must be ticked	You must accept the terms
Login fields	Identifier is an email or valid username; password not empty	Enter your email or username / Enter your password

Code (index.html, HTML + CSS + JavaScript)

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1">
6      <title>Data Validation Demo</title>
7      <style>
8          * { box-sizing: border-box; }
9          body { margin: 0; min-height: 100vh; display: grid; place-items: center; padding: 16px;
10             font-family: "Segoe UI", Roboto, Arial, sans-serif; background: #f1f3f6; color: #1f2937; }
11          .card { width: 440px; max-width: 100%; background: #fff; border-radius: 10px;
12             box-shadow: 0 4px 18px rgba(0,0,0,.10); padding: 24px; }
13          h1 { font-size: 20px; margin: 0 0 14px; }
14          .tabs { display: flex; gap: 8px; margin-bottom: 16px; }
15          .tabs button { flex: 1; padding: 9px; border: 1px solid #d1d5db; background: #f9fafb;
16             border-radius: 6px; cursor: pointer; font-size: 14px; }
17          .tabs button.active { background: #2563eb; border-color: #2563eb; color: #fff; }
18          label { display: block; font-size: 13px; font-weight: 600; margin: 12px 0 4px; }
19          input[type=text], input[type=email], input[type=password], input[type=tel], input[type=date] {
20             width: 100%; padding: 9px 10px; border: 1px solid #d1d5db; border-radius: 6px; font-size: 14px; }
21          input.invalid { border-color: #dc2626; background: #fef2f2; }
22          input.valid { border-color: #16a34a; background: #f0fdf4; }
23          .error { color: #dc2626; font-size: 12px; min-height: 16px; margin-top: 3px; }
24          .check { display: flex; align-items: center; gap: 8px; margin-top: 12px; font-size: 13px; }
25          .btn { width: 100%; margin-top: 14px; padding: 10px; border: 0; border-radius: 6px;
26             background: #2563eb; color: #fff; font-size: 15px; cursor: pointer; }
27          .banner { margin-top: 12px; padding: 10px; border-radius: 6px; font-size: 13px; display: none; }
28          .banner.ok { display: block; background: #dcfce7; color: #166534; }
29          .banner.bad { display: block; background: #fee2e2; color: #991b1b; }
30          .hidden { display: none; }
31      </style>
32  </head>
33  <body>
34      <main class="card">
35          <h1>Account Portal</h1>
36          <div class="tabs">
37              <button id="tab-login" class="active" type="button">Login</button>
38              <button id="tab-signup" type="button">Sign up</button>
39          </div>
40
41          <form id="login-form" novalidate>
42              <label for="l-id">Email or username</label>
43              <input id="l-id" name="id" type="text" placeholder="you@example.com or username">
44              <div class="error" data-for="l-id"></div>
45              <label for="l-pw">Password</label>
46              <input id="l-pw" name="pw" type="password" placeholder="Enter password">
47              <div class="error" data-for="l-pw"></div>
48              <button class="btn" type="submit">Login</button>
49              <div id="login-msg" class="banner"></div>
50          </form>
51
52          <form id="signup-form" class="hidden" novalidate>
53              <label for="s-user">Username</label>
54              <input id="s-user" type="text" placeholder="3-16 letters, digits or _">
55              <div class="error" data-for="s-user"></div>
56              <label for="s-email">Email</label>
57              <input id="s-email" type="email" placeholder="you@example.com">
58              <div class="error" data-for="s-email"></div>
59              <label for="s-phone">Phone</label>
60              <input id="s-phone" type="tel" placeholder="10 digit mobile number">
61              <div class="error" data-for="s-phone"></div>
62              <label for="s-dob">Date of birth</label>
63              <input id="s-dob" type="date">
64              <div class="error" data-for="s-dob"></div>
65              <label for="s-pw">Password</label>
66              <input id="s-pw" type="password" placeholder="Min 8 chars, upper, lower, digit, symbol">
67              <div class="error" data-for="s-pw"></div>
68              <label for="s-pw2">Confirm password</label>
69              <input id="s-pw2" type="password" placeholder="Repeat password">
70              <div class="error" data-for="s-pw2"></div>
71              <div class="check"><input id="s-terms" type="checkbox"><label for="s-terms" style="margin:0;font-
72                 weight:400">I accept the terms and conditions</label></div>
73              <div class="error" data-for="s-terms"></div>
74              <button class="btn" type="submit">Create account</button>
75              <div id="signup-msg" class="banner"></div>
76          </form>
77      </main>
78
79      <script>
80          // ----- validators: each returns "" when valid, otherwise an error message -----

```

```

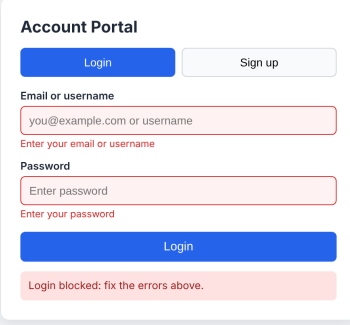
80  const rules = {
81    username: v => /^[A-Za-z0-9_]{3,16}$/ .test(v) ? "" : "Username must be 3-16 letters, digits or underscores",
82    email: v => /^[^s@]+@[^s@]+\.[A-Za-z]{2,}$/ .test(v) ? "" : "Enter a valid email address",
83    phone: v => /^[6-9]\d{9}$/ .test(v) ? "" : "Phone must be 10 digits and start with 6-9",
84    dob: v => {
85      if (!v) return "Date of birth is required";
86      const d = new Date(v), now = new Date();
87      if (isNaN(d) || d > now) return "Enter a valid past date";
88      let age = now.getFullYear() - d.getFullYear();
89      const m = now.getMonth() - d.getMonth();
90      if (m < 0 || (m === 0 && now.getDate() < d.getDate())) age--;
91      return age >= 13 ? "" : "You must be at least 13 years old";
92    },
93    password: v => {
94      if (v.length < 8) return "Password must be at least 8 characters";
95      if (!/[a-z]/.test(v) || !/[A-Z]/.test(v)) return "Use both uppercase and lowercase letters";
96      if (!/\d/.test(v)) return "Add at least one digit";
97      if (!/^[A-Za-z0-9]/.test(v)) return "Add at least one special character";
98      return "";
99    },
100    confirm: (v, all) => v === all.password && v !== "" ? "" : "Passwords do not match",
101    terms: v => v ? "" : "You must accept the terms"
102  };
103
104  const $ = id => document.getElementById(id);
105  function show(id, msg) {
106    const el = $(id);
107    document.querySelector('[data-for="' + id + '"]').textContent = msg;
108    if (el.type !== "checkbox") el.className = msg ? "invalid" : "valid";
109    return !msg;
110  }
111
112  // ----- sign-up -----
113  const signup = $("#signup-form");
114  const readSignup = () => ({
115    username: $("s-user").value.trim(), email: $("s-email").value.trim(),
116    phone: $("s-phone").value.trim(), dob: $("s-dob").value,
117    password: $("s-pw").value, confirm: $("s-pw2").value, terms: $("s-terms").checked
118  });
119  function validateSignup() {
120    const d = readSignup();
121    const results = [
122      show("s-user", rules.username(d.username)),
123      show("s-email", rules.email(d.email)),
124      show("s-phone", rules.phone(d.phone)),
125      show("s-dob", rules.dob(d.dob)),
126      show("s-pw", rules.password(d.password)),
127      show("s-pw2", rules.confirm(d.confirm, d)),
128      show("s-terms", rules.terms(d.terms))
129    ];
130    return results.every(Boolean);
131  }
132  signup.addEventListener("submit", e => {
133    e.preventDefault();
134    const banner = $("#signup-msg");
135    if (validateSignup()) { banner.className = "banner ok"; banner.textContent = "Account created successfully. All inputs passed validation."; }
136    else { banner.className = "banner bad"; banner.textContent = "Please fix the highlighted fields and try again."; }
137  });
138  // live validation after the user leaves a field
139
140  [
    ["s-user", "username"], ["s-email", "email"], ["s-phone", "phone"], ["s-dob", "dob"], ["s-pw", "password"]
  ].forEach(([,id, key]) => {
141    $(id).addEventListener("blur", () => show(id, rules[key]($(id).value.trim())));
142  });
143  // ----- login -----
144  $("#login-form").addEventListener("submit", e => {
145    e.preventDefault();
146    const idv = $("l-id").value.trim(), pw = $("l-pw").value;
147    const okId = show("l-id", !idv ? "Enter your email or username" : (rules.email(idv) === "" || rules.username(idv) === "") ? "" : "Not a valid email or username");
148    const okPw = show("l-pw", pw ? "" : "Enter your password");
149    const banner = $("#login-msg");
150    if (okId && okPw) { banner.className = "banner ok"; banner.textContent = "Input accepted. Sending credentials to the server..."; }
151    else { banner.className = "banner bad"; banner.textContent = "Login blocked: fix the errors above."; }
152  });
153
154  // ----- tabs -----

```

```
155     function tab(which) {
156         $("login-form").classList.toggle("hidden", which !== "login");
157         $("signup-form").classList.toggle("hidden", which !== "signup");
158         $("tab-login").classList.toggle("active", which === "login");
159         $("tab-signup").classList.toggle("active", which === "signup");
160     }
161     $("tab-login").onclick = () => tab("login");
162     $("tab-signup").onclick = () => tab("signup");
163 </script>
164 </body>
165 </html>
166
```

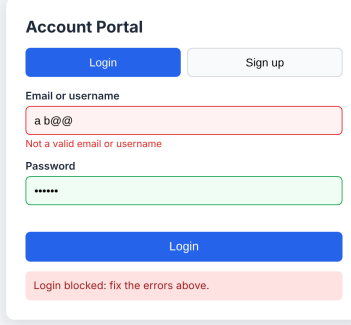
Output

Login form: empty submit, invalid identifier, and accepted input.



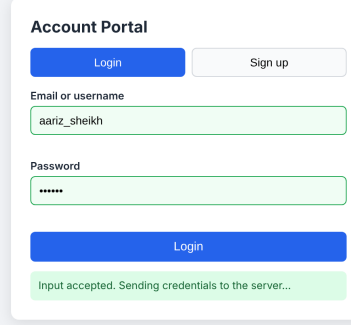
The login form is titled "Account Portal". It has two buttons: "Login" (blue) and "Sign up" (grey). Below the buttons are two input fields: "Email or username" and "Password". The "Email or username" field contains "you@example.com or username" and has a red border with the error message "Enter your email or username" below it. The "Password" field contains "Enter password" and has a red border with the error message "Enter your password" below it. At the bottom, there is a red message box that says "Login blocked: fix the errors above."

Fig. 1: Empty fields blocked



The login form is titled "Account Portal". It has two buttons: "Login" (blue) and "Sign up" (grey). Below the buttons are two input fields: "Email or username" and "Password". The "Email or username" field contains "a b@@" and has a red border with the error message "Not a valid email or username" below it. The "Password" field contains "*****" and has a green border. At the bottom, there is a red message box that says "Login blocked: fix the errors above."

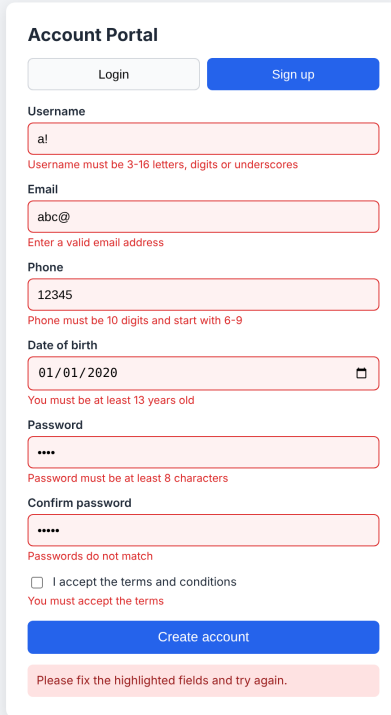
Fig. 2: Invalid email / username rejected



The login form is titled "Account Portal". It has two buttons: "Login" (blue) and "Sign up" (grey). Below the buttons are two input fields: "Email or username" and "Password". The "Email or username" field contains "aariz_sheikh" and has a green border. The "Password" field contains "*****" and has a green border. At the bottom, there is a green message box that says "Input accepted. Sending credentials to the server..."

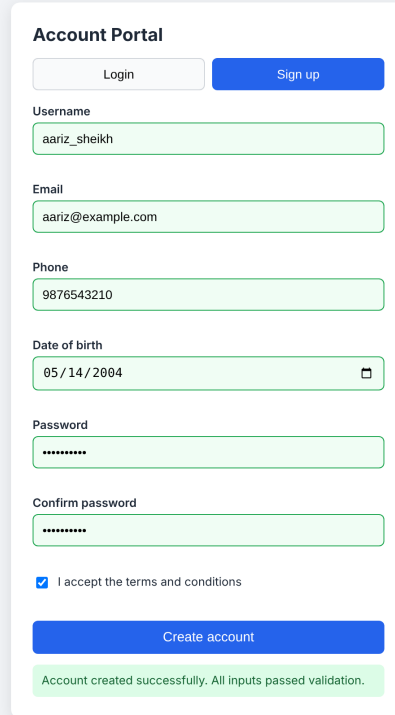
Fig. 3: Valid input accepted

Sign-up form: every field invalid, then every field valid.



The sign-up form is titled "Account Portal". It has two buttons: "Login" (grey) and "Sign up" (blue). Below the buttons are seven input fields: "Username", "Email", "Phone", "Date of birth", "Password", "Confirm password", and a checkbox for "I accept the terms and conditions". The "Username" field contains "al" and has a red border with the error message "Username must be 3-16 letters, digits or underscores" below it. The "Email" field contains "abc@" and has a red border with the error message "Enter a valid email address" below it. The "Phone" field contains "12345" and has a red border with the error message "Phone must be 10 digits and start with 6-9" below it. The "Date of birth" field contains "01/01/2020" and has a red border with the error message "You must be at least 13 years old" below it. The "Password" field contains "****" and has a red border with the error message "Password must be at least 8 characters" below it. The "Confirm password" field contains "*****" and has a red border with the error message "Passwords do not match" below it. The checkbox is unchecked, and there is a red message box that says "You must accept the terms". At the bottom, there is a red message box that says "Please fix the highlighted fields and try again."

Fig. 4: All seven fields fail validation, error under each



The sign-up form is titled "Account Portal". It has two buttons: "Login" (grey) and "Sign up" (blue). Below the buttons are seven input fields: "Username", "Email", "Phone", "Date of birth", "Password", "Confirm password", and a checkbox for "I accept the terms and conditions". The "Username" field contains "aariz_sheikh" and has a green border. The "Email" field contains "aariz@example.com" and has a green border. The "Phone" field contains "9876543210" and has a green border. The "Date of birth" field contains "05/14/2004" and has a green border. The "Password" field contains "*****" and has a green border. The "Confirm password" field contains "*****" and has a green border. The checkbox is checked, and there is a green message box that says "Account created successfully. All inputs passed validation."

Fig. 5: All inputs valid, account created

Conclusion

The program demonstrates data validation by checking each input against presence, format, length, range and consistency rules and displaying a specific error message next to the field that failed. Invalid forms are blocked before submission, and valid forms are accepted. Client-side validation improves user experience, but the same rules must also be enforced on the server, because browser checks can be bypassed.